TT2 Review

Thursday, March 6, 2025 4:38 PM

Bases and Dimensions

Bases : A set of vectors $\{e_1, e_2, \dots e_n\} \in V$ is a basis of V if and only if

① $\{e_1, e_2, \dots e_n\}$ is linearly independent

② $\{e_1, e_2, \dots e_n\}$ spans V

Theorem : Every basis for a given vector space contains the same number of vectors.

Dimensions : The dimension of a vector space, $\dim V$, is the number of vectors in any of its bases.

Proposition : Let V be a finite dimensional vector space with $\dim V = n$. Then,

① A linearly independent set of vectors in V can contain at most n vectors

② A spanning set of V must contain at least n vectors

Theorem : Let $\{v_1, v_2, \dots v_n\} \subset V$ be linearly independent. Then for any vector $v \in V$, $\{v, v_1, v_2, \dots v_n\}$ is linearly independent if and only if $v \notin \operatorname{span}\{v_1, v_2, \dots v_n\}$.

Theorem (Existence of bases) : Let V be a vector space spanned by a finite set of vectors. Then, every linearly independent set of vectors in V can be extended to a basis for V.

Theorem : Let U and W be subspaces of a finite-dimensional vector space V. Then,

① U is finite dimensional and $\dim U \leq \dim V$

② If $U \subseteq W$, then $\dim U \leq \dim W$

③ If $U \subseteq W$ and $\dim U = \dim W$, then $U = W$

} any subspace of a finite-dimensional vector space is finite dimensional

Theorem : Let V be a vector space and $\dim V = n$. Then,

① Any set $\{v_1, v_2, \dots v_n\} \subset V$ that is linearly independent is a basis for V

② Any set $\{v_1, v_2, \dots v_n\} \subset V$ that spans V is a basis of V

Row and column spaces

Suppose $A \in {}^m\mathbb{R}^n$, and $A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} = [c_1\ c_2\ \dots\ c_n]$

Row space : $\operatorname{row} A \triangleq \operatorname{span}\{r_1, r_2, \dots r_m\}$, where $r_i \in \mathbb{R}^n$ are the rows of A

Column space : $\operatorname{col} A \triangleq \operatorname{span}\{c_1, c_2, \dots c_n\}$, where $c_i \in \mathbb{R}^m$ are the columns of A

Important Facts :

↳ $\operatorname{row} A \subseteq \mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ and $\operatorname{col} A \subseteq {}^m\mathbb{R}$ is a subspace of ${}^m\mathbb{R}$.

↳ $\dim \operatorname{row} A = \dim \operatorname{col} A \leq \min\{m, n\}$

↳ $\operatorname{col} A = \{y \mid y = Ax,\ x \in {}^m\mathbb{R}\}$ (col A is the image space of A)

Proposition : Let $A \in {}^m\mathbb{R}^n$, $U \in {}^m\mathbb{R}^m$, $V \in {}^n\mathbb{R}^n$. Then, $\operatorname{row} UA \subseteq \operatorname{row} A$ with equality if U is invertible, and $\operatorname{col} AV \subseteq \operatorname{col} A$ with equality if V is invertible.

proof : Let U have row v_k and A have rows r_k.

$$UA = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{bmatrix} A = \begin{bmatrix} v_1 A \\ v_2 A \\ \vdots \\ v_m A \end{bmatrix}$$

$\therefore$ k^{th} row of UA:

$$r_k' = [v_{k1}\ v_{k2}\ \dots\ v_{kn}]\, A = \sum_{j=1}^{n} v_{kj} r_j,$$

which is a linear combination of rows in A.

$\therefore \operatorname{row} UA = \operatorname{span}\{r_1', r_2', \dots r_m'\} \subseteq \operatorname{row} A$

If U is invertible,

$\operatorname{row} A = \operatorname{row} U^{-1}UA \subseteq \operatorname{row} UA$, so $\operatorname{row} UA = \operatorname{row} A$

Proposition : Let $\{x_1, x_2, \dots x_r\} \subset {}^m\mathbb{R}$ and $U \in {}^m\mathbb{R}^m$ invertible. Then, $\{x_1, x_2, \dots x_m\}$ is linearly independent if and only if $\{Ux_1, Ux_2, \dots Ux_r\}$ is linearly independent.

combining the two propositions, U does not alter linear independence or information if U is invertible.

Lemma I : Let $A \in {}^m\mathbb{R}^n$. Then $\operatorname{row} \tilde{A} = \operatorname{row} A$, where $\tilde{A}$ is RREF of A, and so $\dim \operatorname{row} \tilde{A} = \dim \operatorname{row} A$. Moreover, the nonzero rows of $\tilde{A}$ constitute a basis of $\operatorname{row} A$.

Lemma II : Let $A \in {}^m\mathbb{R}^n$. Then,

① The set of columns with leading 1's $\{\tilde{c}_{j_1}, \tilde{c}_{j_2}, \dots \tilde{c}_{j_r}\}$ of $\tilde{A}$, the RREF, constitutes a basis of $\operatorname{col} \tilde{A}$.

② The corresponding columns $\{c_{j_1}, c_{j_2}, \dots c_{j_r}\}$ of A constitutes a basis of $\operatorname{col} A$.

↳ this is quite different from row, as $\operatorname{row} A = \operatorname{row} \tilde{A}$ but generally $\operatorname{col} A \neq \operatorname{col} \tilde{A}$.

↳ $\dim \operatorname{col} \tilde{A} = \dim \operatorname{col} A$

Theorem : Let $A \in {}^m\mathbb{R}^n$. Then, $\dim \operatorname{row} A = \dim \operatorname{col} A$.

Rank and dimension formula

Let $A \in {}^m\mathbb{R}^n$. The rank of A, $\operatorname{rank} A$, is the common dimension of $\operatorname{row} A$ and $\operatorname{col} A$.

$$\boxed{\operatorname{rank} A = \dim \operatorname{row} A = \dim \operatorname{col} A}$$

Properties :

① If $A \in {}^m\mathbb{R}^n$, then $\operatorname{rank} A = \operatorname{rank} \tilde{A}$

② If $A \in {}^m\mathbb{R}^n$, then $\operatorname{rank} A = \operatorname{rank} A^T$

③ If $A \in {}^m\mathbb{R}^n$, $U \in {}^m\mathbb{R}^m$, and $V \in {}^n\mathbb{R}^n$, then $\operatorname{rank} UA \leq \operatorname{rank} A$ and $\operatorname{rank} AV \leq \operatorname{rank} A$ with equality if U and V are respectively invertible.

Theorem : Let $A \in {}^m\mathbb{R}^n$. Then,

$$\boxed{\dim \operatorname{null} A = n - \operatorname{rank} A}$$

↳ in $Ax = 0$, # of free variables = # of unknowns − # of nonzero rows in A

Number of solutions to $Ax = b$:

① no solution : $b \notin \operatorname{col} A \Rightarrow \operatorname{col} A \subsetneq \operatorname{col}[A|b]$

$$\boxed{\operatorname{rank} A < \operatorname{rank}[A|b]}$$

② unique solution : $b \in \operatorname{col} A \Rightarrow \operatorname{col} A = \operatorname{col}[A|b]$, $\dim \operatorname{null} A = 0$

$$\boxed{\operatorname{rank} A = \operatorname{rank}[A|b] = n}$$

③ infinite solutions : $b \in \operatorname{col} A \Rightarrow \operatorname{col} A = \operatorname{col}[A|b]$, $\dim \operatorname{null} A > 0$

$$\boxed{\operatorname{rank} A = \operatorname{rank}[A|b] < n}$$

given on exam
↓

Theorem : Let $A \in {}^n\mathbb{R}^n$. Then the following statements are equivalent:

① A is invertible

② A has full rank n

③ Rows of A are linearly independent

④ Columns of A are linearly independent

⑤ For $x \in {}^n\mathbb{R}$, $Ax = 0$ implies $x = 0$

⑥ For $z \in {}^n\mathbb{R}$, $z^T A = 0$ implies $z = 0$

Theorem : Let $A \in {}^m\mathbb{R}^n$. Then the following statements are equivalent:

① $\operatorname{rank} A = n$

② columns of A are linearly independent

③ For $x \in {}^n\mathbb{R}$, $Ax = 0 \Rightarrow x = 0$

④ $A^T A$ is invertible

⑤ A has left inverse, ie. $BA = I$ for some $B \in {}^n\mathbb{R}^m$

Theorem : Let $A \in {}^m\mathbb{R}^n$. Then the following statements are equivalent:

① $\operatorname{rank} A = m$

② rows of A are linearly independent

③ for $z \in {}^m\mathbb{R}$, $z^T A = 0$ implies $z = 0$

④ AA^T is invertible

⑤ A has a right inverse, ie. $AB = I$ for some $B \in {}^n\mathbb{R}^m$.

Linear Transformations

The mapping $T : V \to W$ is a linear transformation if and only if

LI. Distribution : $T(u+v) = T(u) + T(v)$ for all $u, v \in V$ (vector addition preserved)

LII. Homogenity : $T(\lambda v) = \lambda T(v)$ for all λ and $v \in V$ (scalar multiplication preserved)

V is called the **domain** and W is the **codomain** or target space of T

Common linear transformations:

- **Trace operator** — $\operatorname{tr} : {}^n\mathbb{R}^n \to \mathbb{R}$ (sum of diagonal entries)
  $\operatorname{tr} A = \sum_{i=1}^{n} a_{ii}$
- **Differential operator** — $D : \mathbb{P}_n \to \mathbb{P}_{n-1}$ (derivative)
- **Identity operator** — $i : V \to V$ (leaves any vector unchanged)
  $i(v) = v$ for any $v \in V$
- **Zero operator** — $z : V \to W$ (takes any vector in V and transforms to zero vector in W)
  $z(v) = 0$, where 0 is zero vector of W

Proposition : Let $T : V \to W$ be a linear transformation. Then,

① $T(0) = 0$

② $T(-v) = -T(v)$ for all $v \in V$

③ for all λ_i and $v_i \in V$,

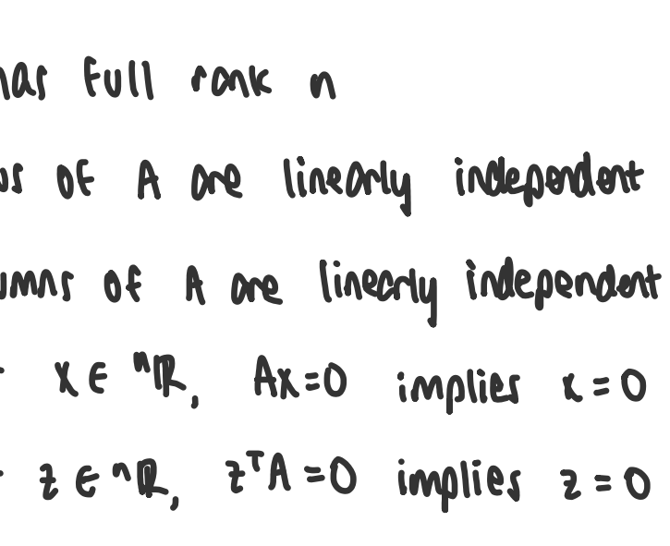
$$T\left(\sum_{i=1}^{k} \lambda_i v_i\right) = \sum_{i=1}^{k} \lambda_i T(v_i)$$

Images and Kernels

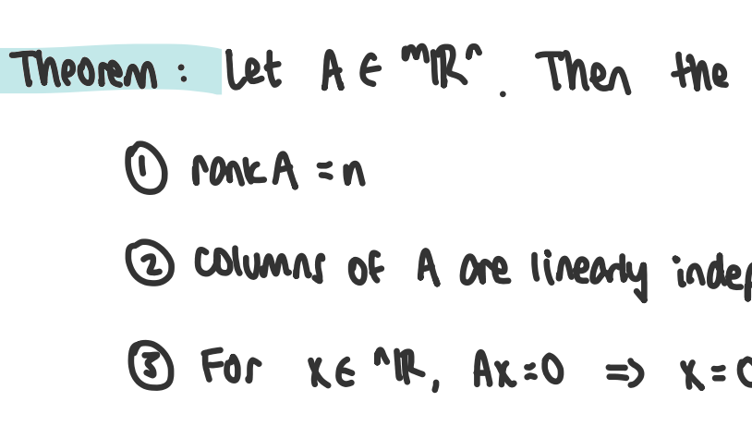
Image : The image of a linear transformation $L : V \to W$ is defined by

$$\boxed{\operatorname{im} T = \{w \mid w = T(v),\ v \in V\}}$$

It is also called the range of T, or $T(V)$.



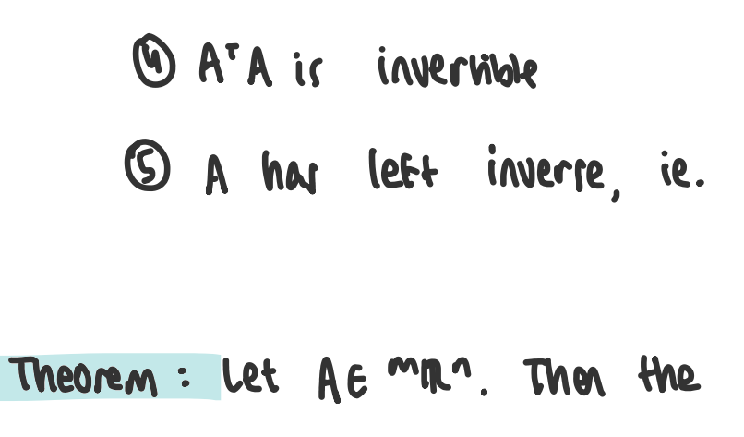
☞ Surjective Mapping



If $\operatorname{im} T = W$, then T is **surjective**

maps V "onto" W

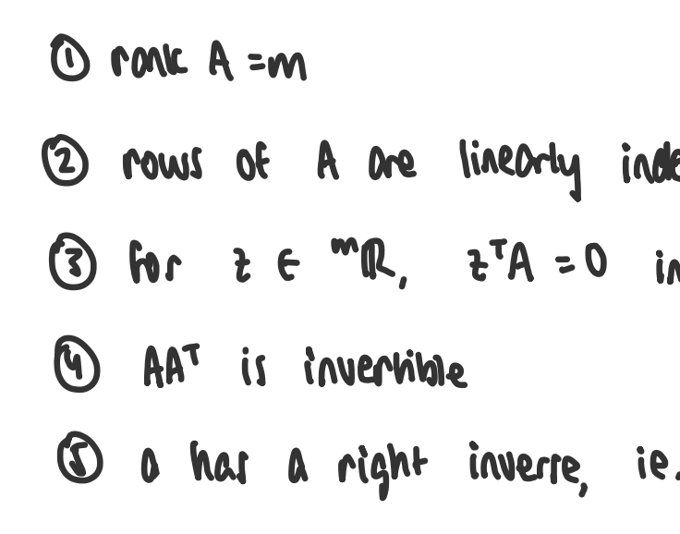
☞ Injective Mapping



If L produces a different $w \in W$ for each $v \in V$, then T is **injective**

"one-to-one" ($T(u) = T(v)$ implies $u = v$)

☞ Bijective Mapping



Both surjective and injective ⇒ **bijective**

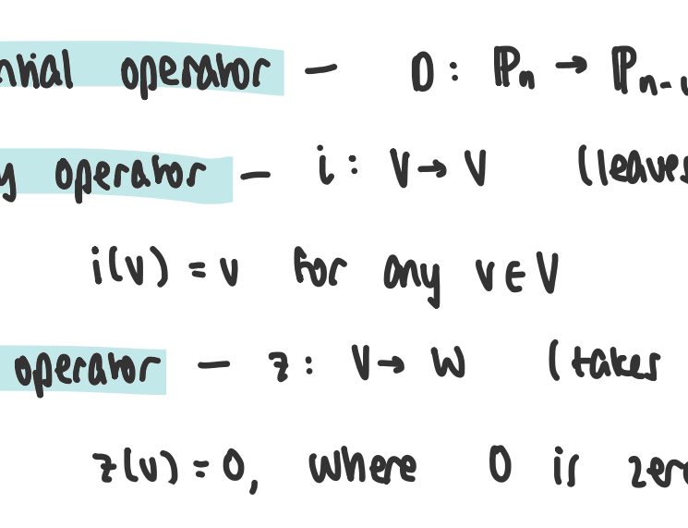
"isomorphic" (same shape)

Kernel : The kernel of a linear transformation $T : V \to W$ is defined as

$$\boxed{\ker T = \{v \in V \mid T(v) = 0\}}$$

(similar to null space of A is x where $Ax = 0$)

$\ker T$ always contains $0 \in V$. If it also contains any other vector, then it cannot be injective (not 1-to-1).



Proposition : Let $T : V \to W$ be a linear transformation. Then, $\operatorname{im} T$ is a subspace of W and $\ker T$ is a subspace of V ($\operatorname{im} T \subseteq W$ and $\ker T \subseteq V$)

Dimension formula : Let $T : V \to W$ be any linear transformation with finite-dimensional $\operatorname{im} T$ and $\ker T$. Then,

$$\boxed{\dim \ker T + \dim \operatorname{im} T = \dim V}$$